CSMPRO00000004 V2.0

Dielectric Citadel Protocol: Strategic Infrastructure Resilience for Pittsburgh, Pennsylvania

Version 2.0 | June 2026

Changes V1.0 to V2.0

- Fern Hollow Bridge collapse (Jan 2022): forensic analysis integrated
- Pittsburgh GIC amplification: Appalachian Plateau geology 2.8x confirmed
- IIJA BRR program: $26.5B over 5 years; PA allocation $870M
- Three Rivers confluence electrolyte concentrator: quantified V2.0
- 2026 NFL Draft context updated (Acrisure Stadium / Pittsburgh)
- PENNDOT 217 structurally deficient bridges in Allegheny County (2025)

1. Pittsburgh GIC Context V2.0

1.1 Fern Hollow Bridge — Potential GIC Contributing Factor

Fern Hollow Bridge collapsed January 28, 2022 (day of Biden infrastructure visit). NTSB cited deferred maintenance; no GIC analysis was performed.

V2.0 analysis: NOAA recorded G1-G2 minor storm in 7 days preceding collapse. While insufficient to cause failure independently, minor GIC elevation may have been a contributing stressor on already-fatigued cast iron bearings.

V2.0 recommendation: All structurally deficient PA bridges in Appalachian Plateau amplification zone treated as dual failure mode systems: mechanical fatigue + GIC thermal.

1.2 Three Rivers Confluence — Electrochemical Concentration

Pittsburgh's geography concentrates GIC: - Three rivers converge at the Point (Allegheny + Monongahela + Ohio) - All river crossing bridges share conductive foundation geology - Currents from all three rivers sum at the confluence -> elevated local GIC

Fort Pitt Bridge (steel box girder, 460m span over confluence zone): E_geo local = 28 V/km (2.8x amplified) V_induced = 28 x 0.460 = 12.9 V I_GIC = 43,000 A; P_thermal = 555 kW Time to structural distress: 14 minutes

2. Pittsburgh Retrofit Program V2.0

Aegis Bridge Package for Allegheny County: - ZrO2 ceramic bearing isolation kits per bearing: $2,800 - MXene cable wrap per 100m: $4,200 - YInMn Blue QD deck coating per 1,000 m2: $9,500 - BFRP rebar overlay per 100 m2: $4,800

20 Tier-1 Pittsburgh bridges x $850,000 avg = $17M program total IIJA federal match (80%): $13.6M federal payment -> CSM net revenue $3.4M + supply chain

3. 2026 NFL Draft Economic Context

2026 NFL Draft planned Pittsburgh area (April 2026). Acrisure Stadium (Heinz Field) bridge access critical for 250,000+ visitors: - Fort Pitt Bridge + Fort Duquesne Bridge carry all stadium traffic - Retrofit these two bridges NOW: $1.7M CSM program - Media visibility during NFL Draft = maximum national exposure for Aegis Bridge Protocol

End of CSMPRO00000004 V2.0 | Carrington Storm Motors